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Agenda & Learning Objectives

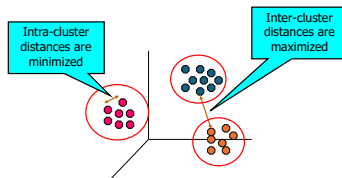
- **Part 1: Core Foundations (Ch. 8)**
 - What is Cluster Analysis? Requirements & Challenges.
 - Partitioning Methods: k-Means, k-Medoids, k-Modes.
 - Hierarchical Methods: Agglomerative vs. Divisive, BIRCH.
 - Density-Based & Grid-Based Methods: DBSCAN, STING, CLIQUE.
 - Evaluation: How to measure "good" clusters (Extrinsic vs. Intrinsic).
- **Part 2: Advanced Techniques (Ch. 9)**
 - Probabilistic Model-Based Clustering (EM Algorithm).
 - High-Dimensional Challenges: The Curse of Dimensionality.
 - Subspace & Biclustering.
 - Clustering Graph & Network Data (SCAN).
 - Semi-Supervised Clustering (Constraints).

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What is Cluster Analysis?

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



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Introduction to Cluster Analysis

Chapter 8:

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Chapter 8: Introduction to Cluster Analysis

- **What is a Cluster?**
 - **Intra-cluster Similarity:** Objects within a cluster are "close" to each other.
 - **Inter-cluster Dissimilarity:** Objects in different clusters are "far apart."
 - **Definition:** The process of grouping a set of physical or abstract objects into classes of *similar* objects.
- **Motivating Example:**
 - **Business:** Segment customers for targeted marketing (e.g., "High-Spend Techies" vs. "Budget Occasional Shoppers").
 - **Biology:** Group genes with similar expression patterns.
 - **Image Processing:** Group pixels to segment objects in a photo.

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Requirements for Clustering in Data Mining

- **Challenges we must solve:**
 - **Scalability:** Must handle millions/billions of objects (e.g., Netflix users).
 - **Mixed Data Types:** Not just numbers. Must handle binary, categorical, ordinal data.
 - **Arbitrary Shapes:** Clusters aren't always spheres. Think of a winding river on a map (snake-shaped cluster).
 - **High Dimensionality:** Curse of dimensionality. Distances become meaningless.
 - **Noise & Outliers:** Algorithms should not be derailed by bad data.
 - **Interpretability:** The business user must understand *why* Group A exists.

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The Problem of Clustering

- Given a set of points, with a notion of distance between points, group the points into some number of *clusters*, so that members of a cluster are in some sense as nearby as possible.
- Clustering is **unsupervised classification**: no predefined classes.
- Formally, Clustering is the process of grouping data points such as intra-cluster distance is minimized and inter-cluster distance is maximized.

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Overview of Clustering Methods (Taxonomy)

- Partitioning Methods**: Construct k partitions. Iteratively relocate objects. (e.g., k-Means)
- Hierarchical Methods**: Build a tree (dendrogram) of clusters. Agglomerative (bottom-up) or Divisive (top-down).
- Density-Based Methods**: Clusters are dense regions separated by sparse regions. (e.g., DBSCAN)
- Grid-Based Methods**: Quantize space into cells; fast processing independent of n . (e.g., STING)

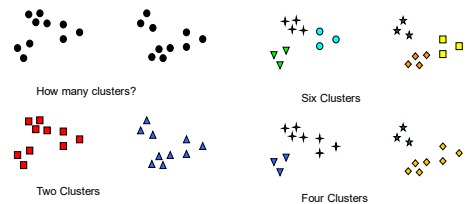
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What is not Cluster Analysis?

- Simple segmentation
 - Dividing students into different registration groups alphabetically, by last name
- Results of a query
 - Groupings are a result of an external specification
 - Clustering is a grouping of objects based on the data
- Supervised classification
 - Have class label information
- Association Analysis
 - Local vs. global connections

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Notion of a Cluster can be Ambiguous



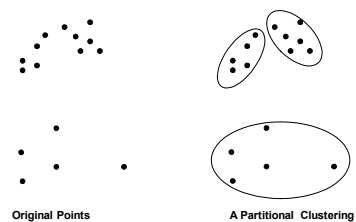
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Types of Clustering

- A clustering is a set of clusters
- Important distinction between hierarchical and partitional sets of clusters
- Partitional Clustering
 - A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset
- Hierarchical clustering
 - A set of nested clusters organized as a hierarchical tree
- Other distinctions – *coming slides*

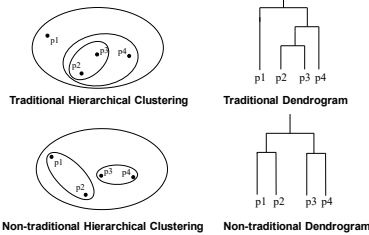
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Partitional Clustering



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Hierarchical Clustering



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Other Distinctions Between Sets of Clusters

- **Exclusive versus non-exclusive**
 - In non-exclusive clusterings, points may belong to multiple clusters.
 - Can represent multiple classes or 'border' points
- **Fuzzy versus non-fuzzy**
 - In fuzzy clustering, a point belongs to every cluster with some weight between 0 and 1
 - Weights must sum to 1
 - Probabilistic clustering has similar characteristics
- **Partial versus complete**
 - In some cases, we only want to cluster some of the data
- **Heterogeneous versus homogeneous**
 - Clusters of widely different sizes, shapes, and densities

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More Types of Clusters

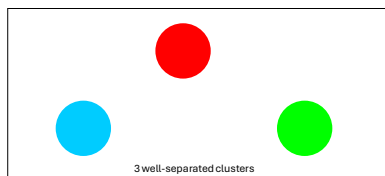
- Well-separated clusters
- Center-based clusters
- Contiguous clusters
- Density-based clusters
- Property or Conceptual
- Described by an Objective Function

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Types of Clusters: Well-Separated

- **Well-Separated Clusters:**
 - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.



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Types of Clusters: Center-Based

- **Center-based**
 - A cluster is a set of objects such that an object in a cluster is closer (more similar) to the "center" of a cluster, than to the center of any other cluster
 - The center of a cluster is often a centroid, the average of all the points in the cluster, or a medoid, the most "representative" point of a cluster



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Types of Clusters: Density-Based

- Density-based

- A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
- Used when the clusters are irregular or intertwined, and when noise and outliers are present.



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Data Structures Used

- Data matrix

$$\begin{bmatrix} x_{11} & \dots & x_{1f} & \dots & x_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ x_{i1} & \dots & x_{if} & \dots & x_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ x_{n1} & \dots & x_{nf} & \dots & x_{np} \end{bmatrix}$$

- Similarity matrix

$$\begin{bmatrix} 0 & & & & \\ d(2,1) & 0 & & & \\ d(3,1) & d(3,2) & 0 & & \\ \vdots & \vdots & \vdots & & \\ d(n,1) & d(n,2) & \dots & \dots & 0 \end{bmatrix}$$

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Clustering Algorithms

- K-means and its variants
- Hierarchical clustering
- Density-based clustering

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Partitioning (Centroid-Based) Algorithms

- Construct a partition of a database D of n objects into a set of k clusters
- Given a k , find a partition of k clusters that optimizes the chosen partitioning criterion
 - k -means (MacQueen'67)
 - Each cluster is represented by the center of the cluster
 - A Euclidean Distance based method, mostly used for interval/ratio scaled data
 - k -medoids
 - Each cluster is represented by one of the objects in the cluster
 - For categorical data

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Partitioning Methods: The k-Means Algorithm

- **Concept:** Centroid-based. The center of a cluster is the *mean* of its points.
- **Algorithm Steps:**
 - **Initialize:** Choose " k " random centers.
 - **Assign:** Assign each point to the *nearest* center.
 - **Update:** Recompute centers as the mean of assigned points.
 - **Repeat** Steps 2 & 3 until convergence (no change).
 - **Objective Function (SSE):** Minimize sum of squared errors.

$$E = \sum_{i=1}^k \sum_{p \in c_i} \text{dist}(p, c_i)^2$$

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K Mean Through an example

A pizza chain wants to open its delivery centres across a city.
What do you think would be the possible challenges?

- They need to analyse the areas from where the pizza is being ordered frequently.
 - They need to understand as to how many pizza stores has to be opened to cover delivery in the area.
 - They need to figure out the locations for the pizza stores within all these areas in order to keep the distance between the store and delivery points minimum.

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K-means Clustering

- Partitional clustering approach
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- Number of clusters, K , must be specified
- The basic algorithm is very simple

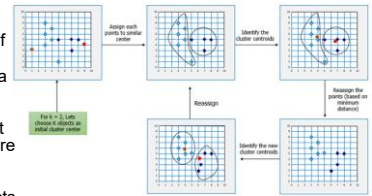
- 1: Select K points as the initial centroids.
- 2: **repeat**
- 3: Form K clusters by assigning all points to the closest centroid.
- 4: Recompute the centroid of each cluster.
- 5: **until** The centroids don't change

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Steps of K Mean Algorithm

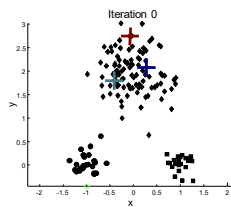
1. Partition of objects into k non-empty subsets
2. Identifying the cluster centroids (mean point) of the current partition.
3. Assigning each point to a specific cluster
4. Compute the distances from each point and allot points to the cluster where the distance from the centroid is minimum.
5. After re-allotting the points, find the centroid of the new cluster formed.



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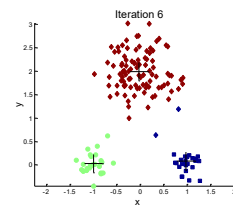
Clustering Example



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Clustering Example



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K-means Clustering – Details

- Initial centroids are often chosen randomly.
 - Clusters produced vary from one run to another.
- The centroid is (typically) the mean of the points in the cluster.
- 'Closeness' is measured by Euclidean distance, cosine similarity, correlation, etc.
- K-means will converge for common similarity measures mentioned above.
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters, I = number of iterations, d = number of attributes

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A Simple example showing the implementation of k-means algorithm (using $K=2$)

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

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Step 1:

Initialization: Randomly we choose following two centroids (k=2) for two clusters.

In this case the 2 centroid are: $m_1=(1.0,1.0)$ and $m_2=(5.0,7.0)$.

Individual	Variable 1	Variable 2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

	Individual	Mean Vector
Group 1	1	(1.0, 1.0)
Group 2	4	(5.0, 7.0)

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Step 2:

• Thus, we obtain two clusters containing:

$\{1,2,3\}$ and $\{4,5,6,7\}$.

• Their new centroids are:

$$d(m_1, 2) = \sqrt{(1.0 - 1.5)^2 + (1.0 - 2.0)^2} = 1.12$$

$$d(m_2, 2) = \sqrt{(5.0 - 1.5)^2 + (7.0 - 2.0)^2} = 6.10$$

$$m_1 = \left(\frac{1}{3}(1.0 + 1.5 + 3.0), \frac{1}{3}(1.0 + 2.0 + 4.0)\right) = (1.83, 2.33)$$

$$m_2 = \left(\frac{1}{4}(5.0 + 3.5 + 4.5 + 3.5), \frac{1}{4}(7.0 + 5.0 + 5.0 + 4.5)\right) = (4.12, 5.38)$$

Individual	Centroid 1	Centroid 2
1	0	7.21
2 (1.5, 2.0)	1.12	6.10
3	3.61	3.61
4	7.21	0
5	4.72	2.5
6	5.31	2.06
7	4.30	2.92

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Step 3:

• Now using these centroids we compute the Euclidean distance of each object, as shown in table.

• Therefore, the new clusters are: $\{1,2\}$ and $\{3,4,5,6,7\}$

• Next centroids are: $m_1=(1.25, 1.5)$ and $m_2 = (3.9, 5.1)$

Individual	Centroid 1	Centroid 2
1	1.57	5.38
2	0.47	4.28
3	2.04	1.78
4	5.64	1.84
5	3.15	0.73
6	3.78	0.54
7	2.74	1.08

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Step 4:

The clusters obtained are:

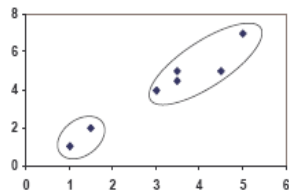
$\{1,2\}$ and $\{3,4,5,6,7\}$

• Therefore, there is no change in the cluster.

• Thus, the algorithm comes to a halt here and final result consist of 2 clusters $\{1,2\}$ and $\{3,4,5,6,7\}$.

Individual	Centroid 1	Centroid 2
1	0.56	5.02
2	0.56	3.92
3	3.05	1.42
4	6.66	2.20
5	4.16	0.41
6	4.78	0.61
7	3.75	0.72

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PLOT

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(with K=3)

Individual	$m_1 = 1$	$m_2 = 2$	$m_3 = 3$	cluster
1	0	1.11	3.61	1
2	1.12	0	2.5	2
3	3.61	2.5	0	3
4	7.21	6.10	3.61	3
5	4.72	3.61	1.12	3
6	5.31	4.24	1.80	3
7	4.30	3.20	0.71	3

clustering with initial centroids (1, 2, 3)

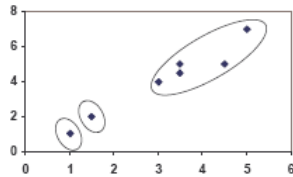
Step 1

Individual	m_1 (1.0, 1.0)	m_2 (1.5, 2.0)	m_3 (3.9, 5.1)	cluster
1	0	1.11	5.02	1
2	1.12	0	3.92	2
3	3.61	2.5	1.42	3
4	7.21	6.10	2.20	3
5	4.72	3.61	0.41	3
6	5.31	4.24	0.61	3
7	4.30	3.20	0.72	3

Step 2

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PLOT



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Real-Life Numerical Example of K-Means Clustering

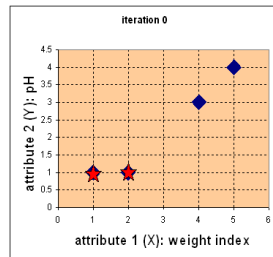
We have 4 medicines as our training data points object and each medicine has 2 attributes. Each attribute represents coordinate of the object. We have to determine which medicines belong to cluster 1 and which medicines belong to the other cluster.

Object	Attribute1 (X): weight index	Attribute 2 (Y): pH
Medicine A	1	1
Medicine B	2	1
Medicine C	4	3
Medicine D	5	4

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Step 1:

- Initial value of centroids :** Suppose we use medicine A and medicine B as the first centroids.
- Let and c_1 and c_2 denote the coordinate of the centroids, then $c_1=(1,1)$ and $c_2=(2,1)$



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- Objects-Centroids distance :** we calculate the distance between cluster centroid to each object. Let us use Euclidean distance, then we have distance matrix at iteration 0 is

$$D^0 = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 1 & 0 & 2.83 & 4.24 \end{bmatrix} \quad \begin{matrix} c_1 = (1,1) & \text{group-1} \\ c_2 = (2,1) & \text{group-2} \end{matrix}$$

- Each column in the distance matrix symbolizes the object.
- The first row of the distance matrix corresponds to the distance of each object to the first centroid and the second row is the distance of each object to the second centroid.
- For example, distance from medicine C = (4, 3) to the first centroid $c_1 = (1, 1)$ is $\sqrt{(4-1)^2 + (3-1)^2} = 3.61$ and its distance to the second centroid is $c_2 = (2, 1)$ is $\sqrt{(4-2)^2 + (3-1)^2} = 2.83$ etc.

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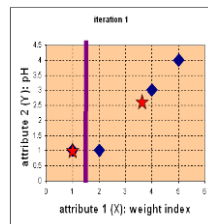
Step 2:

- Objects clustering :** We assign each object based on the minimum distance.
- Medicine A is assigned to group 1, medicine B to group 2, medicine C to group 2 and medicine D to group 2.
- The elements of Group matrix below is 1 if and only if the object is assigned to that group.

$$G^0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad \begin{matrix} \text{group-1} \\ \text{group-2} \end{matrix}$$

A B C D

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- Iteration-1, Objects-Centroids distances :** The next step is to compute the distance of all objects to the new centroids.

- Similar to step 2, we have distance matrix at iteration 1 is

$$D^1 = \begin{bmatrix} 0 & 1 & 3.61 & 5 \\ 3.14 & 2.36 & 0.47 & 1.89 \end{bmatrix} \quad \begin{matrix} c_1 = (1,1) & \text{group-1} \\ c_2 = (\frac{11}{3}, \frac{8}{3}) & \text{group-2} \end{matrix}$$

A B C D

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 1 & 1 & 3 & 4 \end{bmatrix} \quad \begin{matrix} X \\ Y \end{matrix}$$

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• **Iteration-1, Objects**

clustering: Based on the new distance matrix, we move the medicine B to Group 1 while all the other objects remain. The Group matrix is shown below

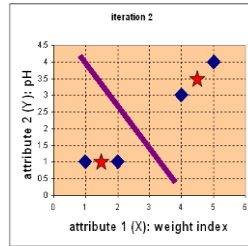
$$G^1 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \text{group -1} \\ \text{group -2} \end{matrix}$$

• **Iteration 2, determine centroids:**

Now we repeat step 4 to calculate the new centroids coordinate based on the clustering of previous iteration. Group 1 and group 2 both has two members, thus the new centroids are and

$$c_1 = \left(\frac{1+2}{2}, \frac{1+1}{2} \right) = (1.5, 1)$$

$$c_2 = \left(\frac{4+5}{2}, \frac{3+4}{2} \right) = (4.5, 3.5)$$



• **Iteration-2, Objects-Centroids distances** : Repeat step 2 again, we have new distance matrix at iteration 2 as

$$D^2 = \begin{bmatrix} 0.5 & 0.5 & 3.20 & 4.61 \\ 4.30 & 3.54 & 0.71 & 0.71 \end{bmatrix} \begin{matrix} c_1 = (1.5, 1) \text{ group -1} \\ c_2 = (4.5, 3.5) \text{ group -2} \end{matrix}$$

	A	B	C	D	
1	2	4	5		X
1	1	3	4		Y

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- **Iteration-2, Objects clustering:** Again, we assign each object based on the minimum distance.

$$G^2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{matrix} \text{group -1} \\ \text{group -2} \end{matrix}$$

- We obtain result that $G^2 = G^1$. Comparing the grouping of last iteration and this iteration reveals that the objects does not move group anymore.
- Thus, the computation of the k-mean clustering has reached its stability and no more iteration is needed..

We get the final grouping as the results as:

Object	Feature1(X): weight index	Feature2 (Y): pH	Group (result)
Medicine A	1	1	1
Medicine B	2	1	1
Medicine C	4	3	2
Medicine D	5	4	2

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k-Means Example & Limitations

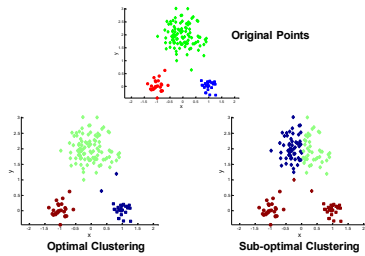
• **Critical Limitations:**

- **Sensitivity to Outliers:** A single extreme value can pull the mean away from the true cluster center (Fig. 8.16 effect).
- **Spherical Clusters Only:** Cannot find complex shapes (concave sets).
- **Fixed "k":** Requires the user to know the number of clusters beforehand.
- **Solution: k-Medoids (PAM)** uses actual data points (medoids) as centers instead of means. This makes it **robust to outliers** but computationally much heavier ($O(k(n-k)^2)$).

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Two different K-means Clusterings



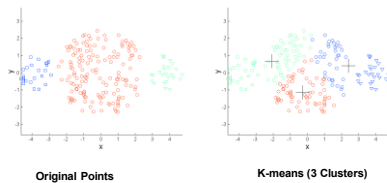
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Limitations of K-means

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.

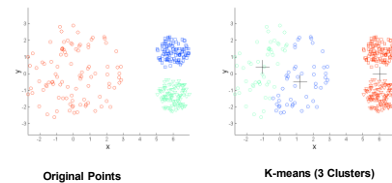
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Limitations of K-means: Differing Sizes



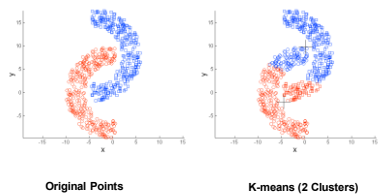
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Limitations of K-means: Differing Density



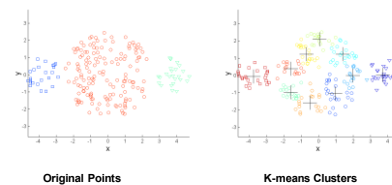
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Limitations of K-means: Non-globular Shapes



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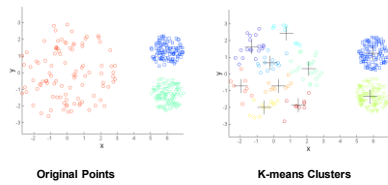
Overcoming K-means Limitations



One solution is to use many clusters.
Find parts of clusters, but need to put together.

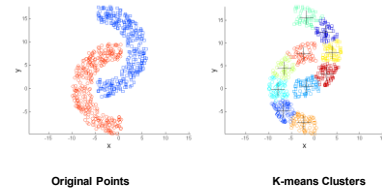
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Overcoming K-means Limitations



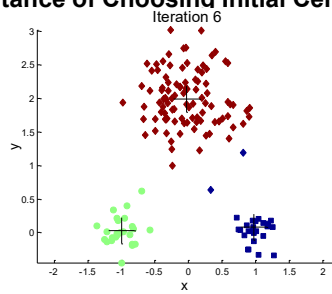
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Overcoming K-means Limitations



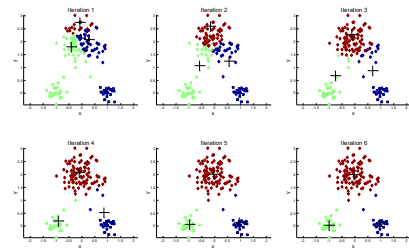
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Importance of Choosing Initial Centroids



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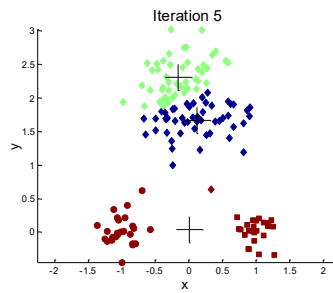
Importance of Choosing Initial Centroids



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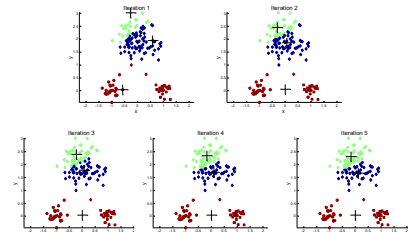
Importance of Choosing Initial Centroids

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Importance of Choosing Initial Centroids ...



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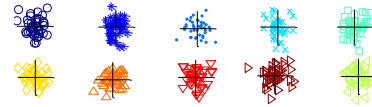
Problems with Selecting Initial Points

- If there are K 'real' clusters then the chance of selecting one centroid from each cluster is small.
- Chance is relatively small when K is large
- If clusters are the same size, n , then

$$p = \frac{\text{number of ways to select one centroid from each cluster}}{\text{number of ways to select } K \text{ centroids}} = \frac{K^n \cdot K^n}{(K \cdot n)^K} = \frac{K!}{K^K}$$

- For example, if $K = 10$, then probability $= 10!/10^{10} = 0.00036$
- Sometimes the initial centroids will readjust themselves in 'right' way, and sometimes they don't
- Consider an example of five pairs of clusters

10 Clusters Example

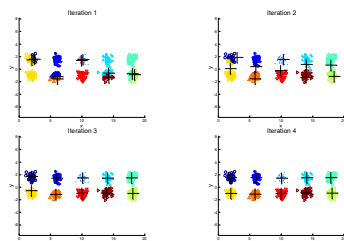


Starting with two initial centroids in one cluster of each pair of clusters

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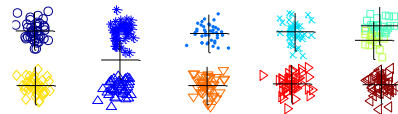
10 Clusters Example



Starting with two initial centroids in one cluster of each pair of clusters

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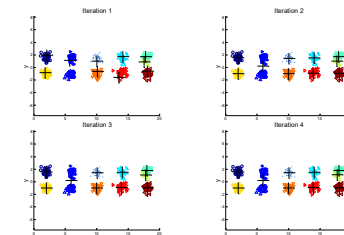
10 Clusters Example



Starting with some pairs of clusters having three initial centroids, while other have only one.

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10 Clusters Example



Starting with some pairs of clusters having three initial centroids, while other have only one.

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Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Sample and use hierarchical clustering to determine initial centroids
- Select more than k initial centroids and then select among these initial centroids
 - Select most widely separated
- Postprocessing
- Generate a larger number of clusters and then perform a hierarchical clustering
- Bisecting K-means
 - Not as susceptible to initialization issues

K-means++

- This approach can be slower than random initialization, but very consistently produces better results in terms of SSE
 - The k-means++ algorithm guarantees an approximation ratio $O(\log k)$ in expectation, where k is the number of centers
- To select a set of initial centroids, C , perform the following
 - Select an initial point at random to be the first centroid
 - For $k - 1$ steps
 - For each of the N points, x_i , $1 \leq i \leq N$, find the minimum squared distance to the currently selected centroids, $C_1, \dots, C_j$, $1 \leq j < k$, i.e., $\min_j d^2(C_j, x_i)$
 - Randomly select a new centroid by choosing a point with probability proportional to $\frac{\min_j d^2(C_j, x_i)}{\sum_i \min_j d^2(C_j, x_i)}$
 - End For

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Variations: k-Modes & Kernel k-Means

- k-Modes (Clustering Categorical Data):**
 - Replace *means* with *modes* (most frequent value).
 - Use a simple matching dissimilarity measure: distance = number of mismatches.
 - Example:** Clustering survey data with attributes like {Married, Single} or {Color: Red, Blue}.
- Kernel k-Means:**
 - Maps data into a higher-dimensional space where non-linear clusters become linearly separable.
 - Uses the "Kernel Trick" (same as SVM) to avoid explicit computation.
 - Example:** Separating an inner ring from an outer ring (target shape). Standard k-Means fails (cuts the ring); Kernel k-Means succeeds.

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